

Cover Letter

Dear Dr. Rus and Dr. Gaban, Guest Editors,

We would like to submit the enclosed manuscript entitled “Summer Heat Stress, Urban Green Infrastructure, and Youth Labour Markets in the European Union: Evidence from a System GMM Approach” for consideration in Sustainability, for the Special Issue “Thermal Mitigation Effects of Green and Blue Infrastructure and Urban Sustainability from an Interdisciplinary Perspective”. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

The Special Issue asks how green and blue infrastructure mitigates urban heat and enhances human well-being, and it explicitly lists the socioeconomic impacts of heatwaves among its themes. The enclosed manuscript answers on the ground where those impacts will be felt longest: the labour-market entry of the young generation. The sectors through which young Europeans start working — tourism, hospitality, agriculture, construction — are the most heat-exposed in the economy, and conditions at entry scar careers for a decade. Using a balanced panel of the 27 EU Member States (2010–2024), cooling degree days, and the green infrastructure endowment of functional urban areas from the Copernicus Urban Atlas, we estimate a dynamic panel model of the youth NEET rate with a two-step System GMM estimator — the methodological register of the recent EU-panel literature in Sustainability.

The highlights of the paper are as follows.

- (1) Summer heat measurably raises youth disengagement. An additional hundred cooling degree days raises the NEET rate by 0.65 percentage points at a zero green endowment ($p < 0.001$), with persistence implying substantially larger long-run effects.
- (2) Urban greenery buffers the shock — the Special Issue’s thermal-mitigation thesis, read in labour-market units. The heat effect declines by 0.0150 percentage points per green-share point ($p = 0.018$), falling from 0.43 at a 15% endowment to statistical insignificance beyond roughly 30% and to essentially zero in the greenest urban systems. Thermal mitigation is, at the macro level, social infrastructure.
- (3) The effect sits on the engagement margin. Heat moves the NEET rate but is undetectable in the youth unemployment rate, which conditions on active search: hot summers push marginal young people out of structured activity altogether. Climate-adaptation dashboards that watch unemployment alone will miss this.
- (4) Disciplined dynamic-panel inference. Two-step System GMM with Windmeijer-corrected standard errors, a deliberately collapsed instrument set (8 instruments for 27 countries), clean Arellano–Bond and Hansen diagnostics (AR(2) $p = 0.876$; Hansen $p = 0.806$), a pre-pandemic robustness estimation, and period demeaning equivalent to full time dummies.

We believe the manuscript fits the Special Issue precisely: it bridges climate science, urban design, and labour economics — the cross-sectoral perspective the call champions — and delivers actionable insights: urban greening under the Nature Restoration Regulation yields youth-employment co-benefits concentrated in the heat-exposed South, and the Youth Guarantee should treat recurring summer shocks as a predictable disengagement risk. We hope the manuscript proves suitable for the Special Issue, and we would be pleased to revise it in light of the reviewers’ comments.

Yours sincerely,

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